

JOBSHEET PRAKTIKUM

Mata Kuliah: Pemrograman Web Lanjut

Pertemuan 3 – Membuat Migration, Model, Relasi & Resource Category

Capaian Pembelajaran

Setelah mengikuti praktikum ini, mahasiswa mampu:

1. Membuat Model dan Migration menggunakan Artisan
 2. Mendesain struktur tabel database
 3. Mengatur \$fillable pada model
 4. Menggunakan \$casts pada Laravel
 5. Membuat relasi antar model (One-to-Many)
 6. Membuat Resource Category di Filament
-

A. Review Singkat

Pada pertemuan sebelumnya kita telah membuat CRUD User menggunakan:

- Filament
- Resource Builder
- Form & Table Schema

Pada pertemuan ini kita akan membuat:

- Model & Migration Category
 - Model & Migration Post
 - Relasi Category → Post
 - Resource Category
-

B. Konsep Dasar

Dalam sistem blog sederhana:

- 1 Category memiliki banyak Post
- Post memiliki 1 Category

Relasi ini disebut One to Many Relationship

C. Langkah Praktikum

◆ Langkah 1 – Membuat Model & Migration Category

Jalankan perintah:

- **php artisan make:model Category -m**

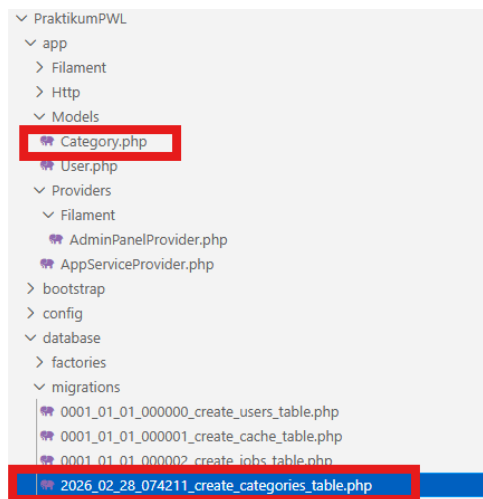
```
PS C:\laragon\www\PraktikumPWL> php artisan make:model Category -m
```

```
INFO Model [C:\laragon\www\PraktikumPWL\app\Models\Category.php] created successfully.
```

```
INFO Migration [C:\laragon\www\PraktikumPWL\database\Migrations\2026_02_28_074211_create_categories_table.php] created successfully.
```

Perintah ini akan membuat:

- **app/Models/Category.php**
- **File migration di folder database/migrations**



◆ Langkah 2 – Mendesain Tabel Categories

Buka file migration Category dan ubah menjadi:

```
PraktikumPWL > database > migrations > 2026_02_28_074211_create_categories_table.php
```

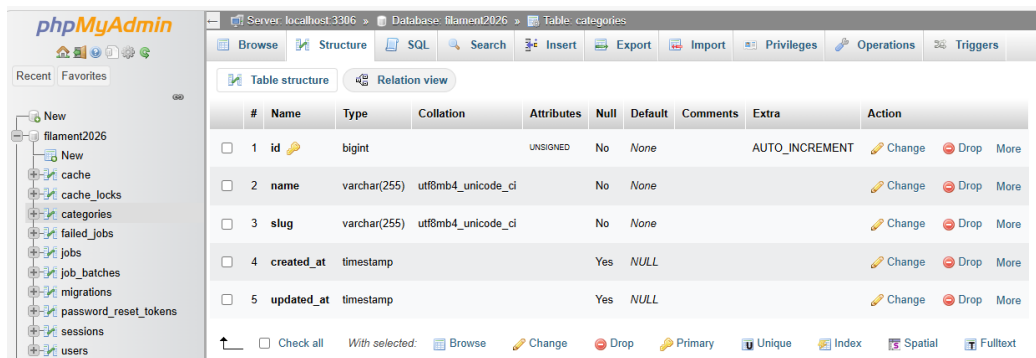
```
1 <?php
2
3 use Illuminate\Database\Migrations\Migration;
4 use Illuminate\Database\Schema\Blueprint;
5 use Illuminate\Support\Facades\Schema;
6
7 return new class extends Migration
8 {
9     /**
10      * Run the migrations.
11      */
12     public function up(): void
13     {
14         Schema::create('categories', function (Blueprint $table) {
15             $table->id();
16             $table->string('name');
17             $table->string('slug');
18             $table->timestamps();
19         });
20     }
21 }
```

Simpan file dan Jalankan migrasi:

- php artisan migrate

```
PS C:\laragon\www\PraktikumPWL> php artisan migrate
INFO: Running migrations.
2026_02_28_074211_create_categories_table ..... 30.02ms DONE
```

Cek database → tabel categories berhasil dibuat.



The screenshot shows the phpMyAdmin interface for a database named 'filament2026'. The 'Table structure' tab is selected for the 'categories' table. The table has 5 columns: 'id' (bigint, UNSIGNED, AUTO_INCREMENT), 'name' (varchar(255), utf8mb4_unicode_ci), 'slug' (varchar(255), utf8mb4_unicode_ci), 'created_at' (timestamp, NULL), and 'updated_at' (timestamp, NULL). Each column has a 'Change', 'Drop', and 'More' action link.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	bigint		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
3	slug	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
4	created_at	timestamp			Yes	NULL			Change Drop More
5	updated_at	timestamp			Yes	NULL			Change Drop More

◆ Langkah 3 – Mengatur Model Category

Buka file:

- app/Models/Category.php

Tambahkan:

- ```
protected $fillable = [
 'name',
 'slug',
];
```

```
PraktikumPWL > app > Models > Category.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Category extends Model
8 {
9 //
10 protected $fillable = [
11 'name',
12 'slug'
13];
14 }
15
```

### 📌 Mengapa perlu fillable?

Agar data dapat disimpan menggunakan mass assignment (digunakan oleh Filament).

---

## ✦ D. Membuat Model & Migration Post

---

### ◆ Langkah 4 – Generate Model Post

- `php artisan make:model Post -m`

```
PS C:\laragon\www\PraktikumPWL> php artisan make:model Post -m
```

```
INFO Model [C:\laragon\www\PraktikumPWL\app\Models\Post.php] created successfully.
```

```
INFO Migration [C:\laragon\www\PraktikumPWL\database\migrations\2026_02_28_075417_create_posts_table.php] created successfully.
```

---

### ◆ Langkah 5 – Mendesain Struktur Tabel Posts

Edit **migration Post** menjadi:

- `$table->string('title');`  
`$table->string('slug');`  
`$table->integer('category_id');`  
`$table->string('color')->nullable();`  
`$table->string('image')->nullable();`  
`$table->text('body')->nullable();`  
`$table->json('tags')->nullable();`  
`$table->boolean('published')->default(false);`  
`$table->date('published_at')->nullable();`

```
PraktikumPWL > database > migrations > 2026_02_28_075417_create_posts_table.php
1 <?php
2
3 use Illuminate\Database\Migrations\Migration;
4 use Illuminate\Database\Schema\Blueprint;
5 use Illuminate\Support\Facades\Schema;
6
7 return new class extends Migration
8 {
9 /**
10 * Run the migrations.
11 */
12 public function up(): void
13 {
14 Schema::create('posts', function (Blueprint $table) {
15 $table->id();
16 $table->string('title');
17 $table->string('slug');
18 $table->integer('category_id');
19 $table->string('color')->nullable();
20 $table->string('image')->nullable();
21 $table->text('body')->nullable();
22 $table->json('tags')->nullable();
23 $table->boolean('published')->default(false);
24 $table->date('published_at')->nullable();
25 $table->timestamps();
26 });
27 }
28 }
```

Jalankan:

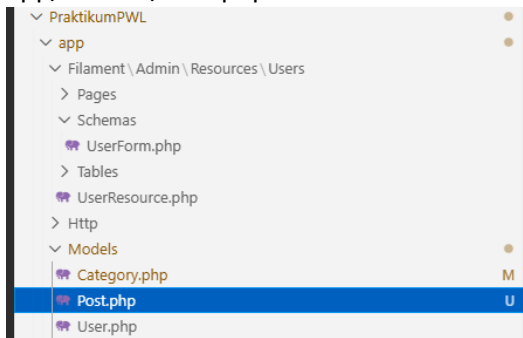
- `php artisan migrate`

```
PS C:\laragon\www\PraktikumPWL> php artisan migrate
[INFO] Running migrations.
2026_02_28_075417_create_posts_table 45.46ms DONE
```

## 📌 E. Mengatur Model Post

Buka:

- app/Models/Post.php



Tambahkan \$fillable:

- protected \$fillable = [  
    'title',  
    'slug',  
    'category\_id',  
    'color',  
    'image',  
    'body',  
    'tags',  
    'published',  
    'published\_at',  
];

```
PraktikumPWL > app > Models > Post.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Post extends Model
8 {
9 //
10 protected $fillable = [
11 'title',
12 'slug',
13 'category_id',
14 'color',
15 'image',
16 'body',
17 'tags',
18 'published',
19 'published_at',
20];
21
22 }
23
```

---

### ◆ Menggunakan Casting Laravel

Karena beberapa field memiliki tipe khusus, tambahkan: ( **pada models/Post.php** )

- protected \$casts = [  
    'tags' => 'array',  
    'published' => 'boolean',  
    'published\_at' => 'date',  
];

```
PraktikumPWL > app > Models > Post.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Post extends Model
8 {
9 //
10 protected $fillable = [
11 'title',
12 'slug',
13 'category_id',
14 'color',
15 'image',
16 'body',
17 'tags',
18 'published',
19 'published_at',
20];
21 protected $casts = [
22 'tags' => 'array',
23 'published' => 'boolean',
24 'published_at' => 'date',
25];
26
27 }
28
```

### ✦ Fungsi casting:

- JSON → array otomatis
  - Boolean → true/false
  - Date → otomatis menjadi Carbon instance
-

◆ Menambahkan Relasi ( pada model/Post.php )

Tambahkan relasi ke Category:

- public function category()  
{  
    return \$this->belongsTo(Category::class);  
}

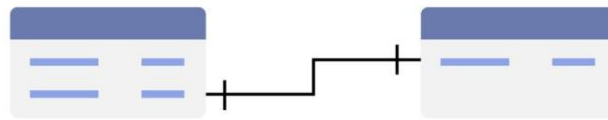
```
PraktikumPWL > app > Models > Post.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Post extends Model
8 {
9 //
10 protected $fillable = [
11 'title',
12 'slug',
13 'category_id',
14 'color',
15 'image',
16 'body',
17 'tags',
18 'published',
19 'published_at',
20];
21 protected $casts = [
22 'tags' => 'array',
23 'published' => 'boolean',
24 'published_at' => 'date',
25];
26
27 public function category()
28 {
29 return $this->belongsTo(Category::class);
30 }
31
32 }
```

Artinya:

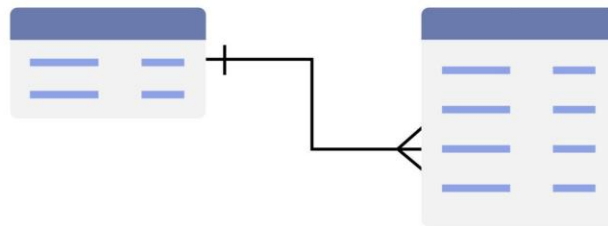
- Setiap Post milik satu Category
-

## ✦ F. Diagram Relasi Database

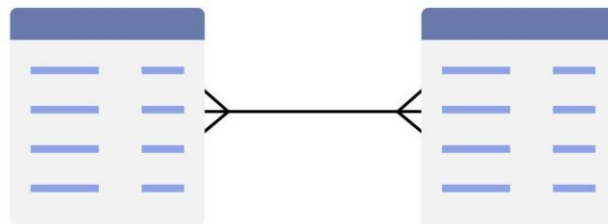
One to one relationship



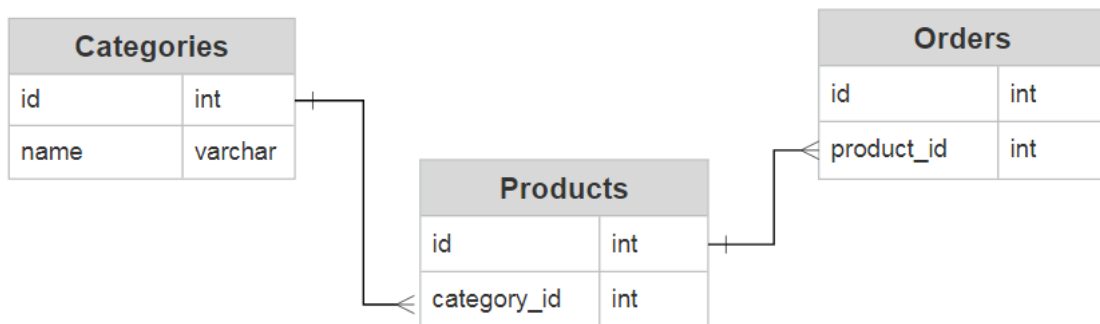
One to many relationship

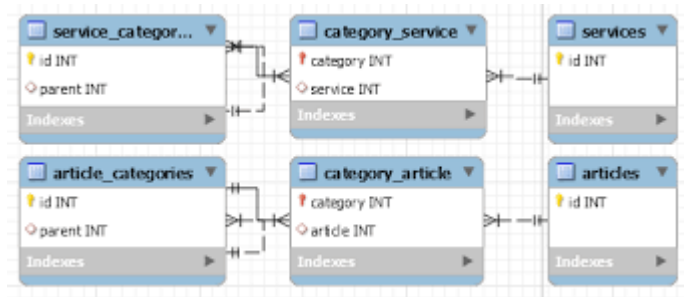


Many to many relationship



## Laravel - Has Many Through Relationship





## ✦ G. Membuat Resource Category di Filament

Karena kita akan menggunakan Category untuk relasi Post, maka kita buat resource-nya terlebih dahulu.

- **php artisan make:filament-resource Category**

Isi:

- Attribute utama → name
- Generate read-only → No
- Generate dari database → No

```
PS C:\laragon\www\PraktikumPWL> php artisan make:filament-resource Category
```

The "title attribute" is used to label each record in the UI.

You can leave this blank if records do not have a title.

What is the title attribute for this model?

> name

Would you like to generate a read-only view page for the resource? (yes/no) [no]

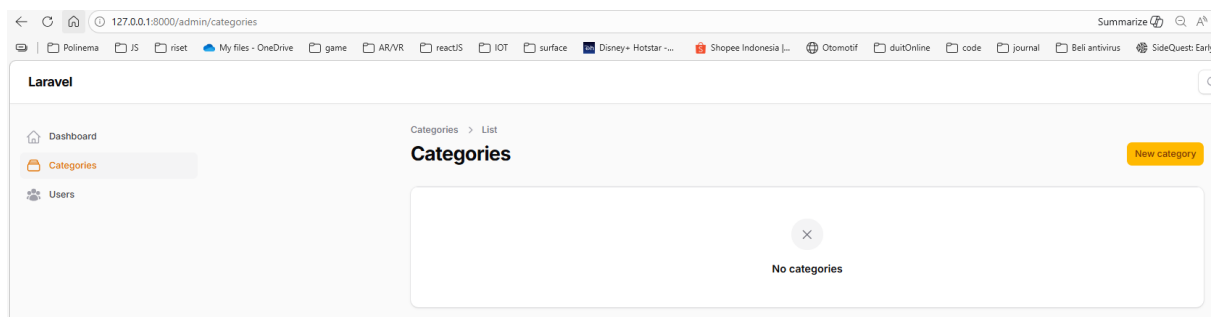
> no

Should the configuration be generated from the current database columns? (yes/no) [no]

> no

**INFO** Filament resource [App\Filament\Admin\Resources\Categories\CategoryResource] created successfully.

Jalankan browser maka akan muncul pada sidebar categories :



## ◆ Edit Form Category

Buka:

- CategoryForm.php

Tambahkan:

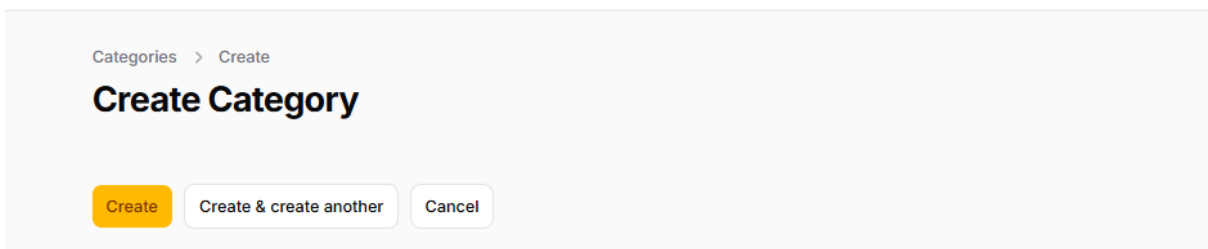
- `TextInput::make('name')`  
`->required(),`

`TextInput::make('slug')`  
`->required(),`

```
PraktikumPWL > app > Filament > Admin > Resources > Categories > Schemas > CategoryForm.php
1 <?php
2
3 namespace App\Filament\Admin\Resources\Categories\Schemas;
4
5 use Filament\Schemas\Schema;
6 use Filament\Forms\Components\TextInput;
7
8 class CategoryForm
9 {
10
11 public static function configure(Schema $schema): Schema
12 {
13 return $schema
14 ->components([
15 //
16 TextInput::make('name')
17 ->required(),
18 TextInput::make('slug')
19 ->required(),
20]);
21 }
22 }
23
24
```

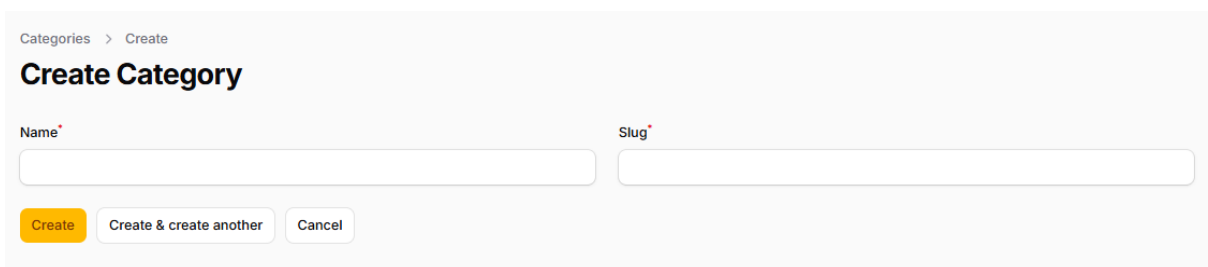
Jalankan browser :

Sebelum



The screenshot shows the 'Create Category' form in a web browser. The breadcrumb is 'Categories > Create'. The title is 'Create Category'. Below the title, there are three buttons: 'Create' (highlighted in orange), 'Create & create another', and 'Cancel'.

Sesudah terdapat form input

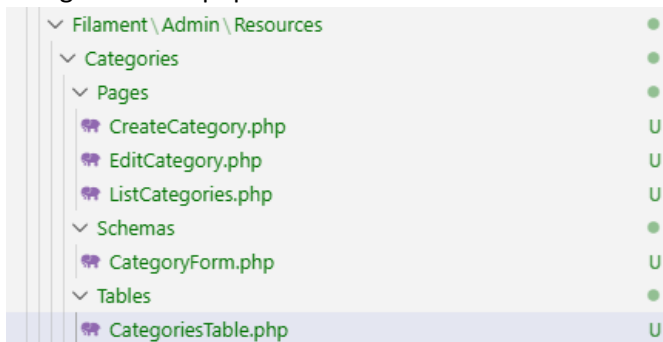


The screenshot shows the 'Create Category' form after adding input fields. The breadcrumb is 'Categories > Create'. The title is 'Create Category'. Below the title, there are two input fields: 'Name' and 'Slug', both with red asterisks indicating they are required. Below the input fields, there are three buttons: 'Create' (highlighted in orange), 'Create & create another', and 'Cancel'.

## ◆ Edit Table Category

Buka:

- CategoriesTable.php



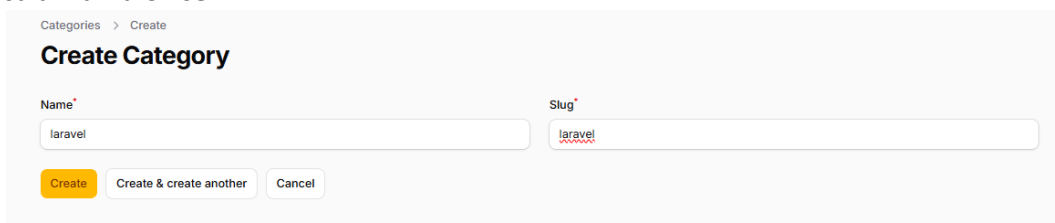
Tambahkan:

- TextColumn::make('name'),  
TextColumn::make('slug'),

PraktikumPWL > app > Filament > Admin > Resources > Categories > Tables > CategoriesTable.php

```
1 <?php
2
3 namespace App\Filament\Admin\Resources\Categories\Tables;
4
5 use Filament\Actions\BulkActionGroup;
6 use Filament\Actions\DeleteBulkAction;
7 use Filament\Actions\EditAction;
8 use Filament\Tables\Table;
9 use Filament\Tables\Columns\TextColumn;
10
11 class CategoriesTable
12 {
13 public static function configure(Table $table): Table
14 {
15 return $table
16 ->columns([
17 //
18 TextColumn::make('name'),
19 TextColumn::make('slug'),
20])
21 ->filters([
22 //
23])
24 ->recordActions([
25 EditAction::make(),
26])
27 ->toolbarActions([
28 BulkActionGroup::make([
29 DeleteBulkAction::make(),
30]),
31]),
32];
33 }
34 }
35
```

Jalankan browser



Categories > List

Categories

New category

| <input type="checkbox"/> Name    | Slug    |                      |
|----------------------------------|---------|----------------------|
| <input type="checkbox"/> laravel | laravel | <a href="#">Edit</a> |

Showing 1 result

Per page10

Note : data berhasil ditampilkan

## H. Tampilan Resource Category

Courses > Filament Example - Nested resources > Lessons > List

Lessons

New lesson

| <div> <div>Q Search</div> <div></div> </div>            |                      |                        |
|---------------------------------------------------------|----------------------|------------------------|
| <input type="checkbox"/> Title                          |                      |                        |
| <input type="checkbox"/> Lesson 1 - Preparations        | <a href="#">Edit</a> | <a href="#">Delete</a> |
| <input type="checkbox"/> Lesson 2 - Building Database   | <a href="#">Edit</a> | <a href="#">Delete</a> |
| <input type="checkbox"/> Lesson 3 - Building Resources  | <a href="#">Edit</a> | <a href="#">Delete</a> |
| <input type="checkbox"/> Lesson 4 - Linking resources   | <a href="#">Edit</a> | <a href="#">Delete</a> |
| <input type="checkbox"/> Lesson 5 - Building Traits     | <a href="#">Edit</a> | <a href="#">Delete</a> |
| <input type="checkbox"/> Lesson 6 - Covering edge cases | <a href="#">Edit</a> | <a href="#">Delete</a> |

Showing 1 to 6 of 6 results

Per page10

Members

Name \*

Taylor Otwell

Role \*

Owner

▼

🗑️

Name \*

Mohamed Said

Role \*

Administrator

⬆️

▼

🗑️

Name \*

Dries Vints

Role \*

Member

⬆️

🗑️

+ Add to members

Filament Menu Manager

Search

🔔

DU

Dashboard

Menu Manager

Shop

Products

Customers

Orders

Categories

Brands

Blog

Posts

Categories

Authors

Menu Manager

Location\* Header

Custom Link

Title\*

URL\*

Target\*

Same Tab

Add Menu Item

Fixed Link

Products

You can edit menu items using the navigation buttons on the right side of the menu item. Click the edit button to see more features.

Products

Analytics

Engagement

Security

Integrations

Automations

Features

Marketplace

Company

About Us

## I. Hasil yang Diharapkan

Mahasiswa berhasil:

- ✓ Membuat tabel categories & posts
  - ✓ Mengatur fillable dan casts
  - ✓ Membuat relasi belongsTo
  - ✓ Membuat CRUD Category
  - ✓ Menampilkan data di tabel
- 

## J. Analisis & Diskusi

1. Mengapa kita perlu \$fillable?
  2. Apa fungsi \$casts pada Laravel?
  3. Apa perbedaan integer biasa dengan foreign key?
  4. Bagaimana jika category dihapus tetapi masih ada post?
- 

## K. Tugas Praktikum

1. Tambahkan minimal 3 kategori berbeda
  2. Tambahkan validasi slug harus unik
  3. Ubah category\_id menjadi foreign key
  4. Screenshot:
    - Struktur tabel di database
    - Form Category
    - List Category
  5. Buat laporan PDF
- 

## L. Kesimpulan

Pada pertemuan ini mahasiswa telah memahami:

- Cara membuat Migration & Model
- Mendesain struktur database
- Menggunakan Casting Laravel
- Membuat relasi antar model
- Membuat Resource Category